

Disproof of Conjecture 00000002831

A stochastic HMM image is not an algebraic set

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Claim addressed. The conjecture states that the image of the transition–emission parameter space in observed distributions is an algebraic variety. With stochastic parameters, the image itself need not be an algebraic set. A one-hidden-state, two-symbol hidden Markov model already gives a counterexample. The distinction between the image and its Zariski closure is essential.

Proposition 1. *The one-observation distribution image of this HMM is the segment*

$$S = \{(t, 1 - t) : 0 \leq t \leq 1\} \subset \mathbb{R}^2.$$

There is no family of real polynomials whose common zero locus is S .

Proof. With one hidden state, the initial distribution and the transition matrix are necessarily (1) and (1). A stochastic emission row has the form $(t, 1 - t)$, where $0 \leq t \leq 1$. The one-observation probabilities are

$$p_j = \sum_{i=0}^0 \pi_i B_{ij} = B_{0j}.$$

Thus the image is exactly the displayed segment. In particular, it contains $(t, 1 - t)$ for every $t \in [0, 1]$, and every point in the image has nonnegative coordinates.

Suppose a family $\mathcal{F} \subset \mathbb{R}[X, Y]$ has common zero locus S . For any $f \in \mathcal{F}$, define the ordinary polynomial

$$q_f(T) = f(T, 1 - T).$$

It vanishes at every $T \in [0, 1]$. A nonzero polynomial in one variable has finitely many roots; this interval is infinite. Therefore q_f is identically zero. In particular, $f(2, -1) = q_f(2) = 0$ for every $f \in \mathcal{F}$. Their common zero locus must contain $(2, -1)$, contrary to its negative second coordinate. This proves the proposition. $\square$

Scope. The counterexample uses normalized, nonnegative probability parameters, exactly as required for a stochastic HMM and observed probability distributions. It addresses the image itself, as the source states. Its Zariski closure is the affine line $X + Y = 1$, which is algebraic; the assertion with “Zariski closure of the image” would evade this counterexample. The argument does not address the separate statement about invariant degrees or tree parameters. It also rules out a complex polynomial zero-locus description of this real segment: restricting such equations to real points and separating real and imaginary parts would give a real polynomial description.

Lean correspondence and reproduction

The structure `HMM` stores an initial distribution, transition matrix and emission matrix with their nonnegativity and row-sum conditions. The hidden-state type is `Fin 1`, and the observation type

is `Fin 2`. The observation map is the actual finite sum of initial probabilities times emission probabilities. Its set-theoretic range is `modelImage`.

Lean constructs a valid HMM for every $t \in [0, 1]$, proves that its observation vector is $(t, 1 - t)$, and proves every observation coordinate is nonnegative. It then uses genuine `MvPolynomial` evaluation and substitution into `Polynomial` to prove that every polynomial vanishing on the segment vanishes on the whole line. Infinitude of the real interval and the finite-root theorem are used directly.

The final theorem `conjecture_false` negates the existence of any family of real polynomial equations defining the image. This is stronger than merely showing the absence of a single defining equation. The Lean contradiction only needs the inclusion of the segment in the image and exclusion of $(2, -1)$; these are proved from the stochastic model, without assuming the image formula.

Inside `lean/`, run `lake build`, followed by

```
lake env lean -DwarningAsError=true Main.lean.
```

Lean is pinned to 4.19.0 and Mathlib to

```
c44e0c8ee63ca166450922a373c7409c5d26b00b.
```

All transitive dependencies are pinned. No numerical approximation is needed. The proof contains no gaps, custom axioms or native computation shortcuts. Actual fresh-build logs, axiom dependencies and hashes are included under `verification/`. Author and delegated-agent review records are separate; no external independent review is claimed.

References

- [1] The Last Math Competition, Conjecture 00000002831 (original bilingual source).